

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000483 - Software Design

DEGREE PROGRAMME

10AM - Master Universitario En Ingenieria Del Software

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Prior knowledge recommended to take the subject.....	2
4. Skills and learning outcomes	3
5. Brief description of the subject and syllabus.....	4
6. Schedule.....	6
7. Activities and assessment criteria.....	8
8. Teaching resources.....	12
9. Other information.....	12

1. Description

1.1. Subject details

Name of the subject	103000483 - Software Design
No of credits	4 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	10AM - Master Universitario en Ingeniería del Software
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Jaime Ramirez Rodriguez (Subject coordinator)	5112	jaime.ramirez@upm.es	Sin horario. The tutoring table can be found at: https://upm365-my.sharepoint.com/:x:/g/personal/m_jimenez_upm_es/EUOWGvHaU4JLmXdlPlxt2eQBQZ_wNYIFNXFGLzdiSm0Nqw?e=7Vfgai

Angelica De Antonio Jimenez	5108	angelica.deantonio@upm.es	M - 12:00 - 14:00 Tu - 17:00 - 19:30 W - 10:30 - 12:00 The tutoring table can be found at: https://upm365-my.sharepoint.com/:x:/g/personal/m_jimenez_upm_es/EUOWGvHaU4JLmXdIplxt2eQBQZ_wNYIFNXFGLzdiSm0Nqw?e=7Vfgai
--------------------------------	------	---------------------------	---

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Object Oriented Programming

4. Skills and learning outcomes *

4.1. Skills to be learned

CE12 - Concebir y realizar el diseño de los sistemas software asegurando atributos relevantes de calidad.

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio (RD)

CG10 - Capacidad de pensamiento creativo con el objetivo de desarrollar enfoques y métodos nuevos y originales

CG11 - Integración del conocimiento a partir de disciplinas diferentes, así como el manejo de la complejidad

CG14 - Conocimiento y comprensión de la informática necesaria para la creación de modelos de información, y de los sistemas y procesos complejos

CG18 - Capacidad de trabajar y comunicarse también en contextos internacionales

CG3 - Que los estudiantes sepan comunicar sus conclusiones y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades (RD)

4.2. Learning outcomes

RA16 - The student will be able to evaluate any software system design.

RA14 - The student will be able to design a software system according to requirements, restrictions, quality standards, and developer criteria

RA15 - The student will be able to document each new design.

RA23 - Time organization capability SC13, SC14 K

RA22 - Observing capability SC13, SC14, CG10 C

RA21 - Listening capability SC13, SC14, CG10 A

RA27 - Negotiation skill SC13, SC14, CG18 C

RA25 - Communication skills in public SC13, SC14, CG3, CG18 S

RA24 - Conflict solving capability SC13, SC14, CG18 C

RA26 - Group work skill SC13, SC14, CG17 A

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

This subject addresses the usage of object-oriented principles to model and design complex systems. It involves analyzing the problem domain, representing it using objects and their interactions, and then designing a modular solution. Also, this subject covers Object-Oriented Modeling Principles such as abstraction, encapsulation, decomposition, and generalization, and different types of inheritance and dependencies. Design Guidelines for modularity, separation of concerns, information hiding, and conceptual integrity are taught to create a flexible, reusable, maintainable design.

A significant part of the course is dedicated to Design Patterns, reusable solutions to common problems in software design, and Unified Modeling Language (UML), a standardized notation for creating diagrams that represent different aspects of a software system.

5.2. Syllabus

1. Introduction to the development process
2. Requirements as entry point to the design
3. Domain modelling
4. High-level Design
5. Low-level Design
6. Design Principles and Patterns

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction to the development process Duration: 02:00 Lecture			
2	Domain modelling Duration: 02:00 Lecture			
3	High-level design Duration: 02:00 Lecture			
4	Workshop of high-level design Duration: 02:00 Problem-solving class			Practical assignment 1 Group work Progressive assessment and Global Examination Not Presential Duration: 08:00
5	Low-level design Duration: 02:00 Lecture			
6	Workshop of high-level design Duration: 01:00 Problem-solving class Low-level design Duration: 01:00 Lecture			Practical assignment 2 Group work Progressive assessment and Global Examination Not Presential Duration: 08:00
7	Low-level design Duration: 02:00 Lecture			
8	Design Principles and Patterns Duration: 02:00 Lecture			
9	Workshop of low-level design Duration: 02:00 Problem-solving class			Practical assignment 3 Group work Progressive assessment and Global Examination Not Presential Duration: 08:00
10	Design Principles and Patterns Duration: 02:00 Lecture			
11	Design Principles and Patterns Duration: 02:00 Lecture			

12	Design Principles and Patterns Duration: 02:00 Lecture			Practical assignment 4 Group work Progressive assessment and Global Examination Not Presential Duration: 08:00
13	Design Principles and Patterns Duration: 02:00 Lecture			
14	Design Principles and Patterns Duration: 02:00 Lecture			
15	Student presentations of the assignment Duration: 03:00 Additional activities			Student presentations of the assignment Group presentation in the lab Progressive assessment and Global Examination Presential Duration: 03:00
16	Exam Duration: 02:00 Additional activities			Exam Written test Progressive assessment and Global Examination Presential Duration: 02:00 Practical assignment 5 Group work Progressive assessment and Global Examination Not Presential Duration: 16:00
17				

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
4	Practical assignment 1	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
6	Practical assignment 2	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
9	Practical assignment 3	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
12	Practical assignment 4	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
15	Student presentations of the assignment	Group presentation in the lab	Face-to-face	03:00	5%	4 / 10	CG3
16	Exam	Written test	Face-to-face	02:00	20%	4 / 10	CE12 CG11
16	Practical assignment 5	Group work	No Presential	16:00	35%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
4	Practical assignment 1	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
6	Practical assignment 2	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
9	Practical assignment 3	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
12	Practical assignment 4	Group work	No Presential	08:00	10%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1
15	Student presentations of the assignment	Group presentation in the lab	Face-to-face	03:00	5%	4 / 10	CG3
16	Exam	Written test	Face-to-face	02:00	20%	4 / 10	CE12 CG11
16	Practical assignment 5	Group work	No Presential	16:00	35%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
-------------	----------	------	----------	--------	---------------	------------------

Exam	Written test	Face-to-face	02:00	20%	4 / 10	CE12 CG11
Practical assignment	Individual work	Face-to-face	10:00	80%	4 / 10	CE12 CG10 CG11 CG14 CG18 CG1 CG3

7.2. Assessment criteria

1. Regular Period

1.1. Distributed evaluation

Throughout the semester, in order to pass the course, the student will have to do the following assessment activities:

- Practical assignments: they will allow the student to apply the concepts, techniques, and principles explained in the classroom.
- Final exam: the student will have to do an exam where he/she will have to show that he/she has acquired the concepts explained in the classroom.

The final grade (FG) of the subject will be calculated from the grades of the practical assignments (PA1G, PA2G, PA3G, PA4G and PA5G) and the exam grade (EG) by means of the following formula:

$FG = 0.1 \cdot (PA1G + PA3G + PA4G) + 0.15 \cdot PA2G + 0.30 \cdot PA5G + 0.2 \cdot EG$, if $EG \geq 4$ and $Mean(PAXG) \geq 4$

$FG = 0$, otherwise

Where all the grades take values between 0 and 10.

1.2. Global evaluation

Due to the proximity of final assignment deadline, it will not be able to be recovered in the global evaluation. Additionally, as all the assignments are related to the same project and the solutions of the previous assignments will be explained in the classroom, these assignments will not be able to be recovered either.

The only assessment activity that will be able to be recovered will be the final exam.

To calculate the final grade, the same formula of the distributed evaluation will be employed.

2. Extraordinary Period

When failed, in this extraordinary period the final grade will be obtained from the grade of a practical assignment (80%) and an exam (20%).

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Moodle	Bibliography	It contains or addresses the fundamental literature

9. Other information

9.1. Other information about the subject

In this course, the following innovative teaching methodology is applied:

- **Project-Based Learning:** it is implemented by engaging students teams in the development of a complete project. Professors propose a problem statement for the project with several deliverables. The project is then developed in several phases, each with a formal deliverable that represents a key part of the solution. For every deliverable, the professor discusses a possible solution with the students in the classroom and provides feedback for the students' solutions. At the end of the course, each team proposes a possible extension of the project in an oral presentation and discusses possible improvements with the professors, demonstrating not only their technical work but also their ability to communicate and reflect on the process.